Concept Review

Servo Motors

Why use Servo Motors?

For many mechatronic applications, it is necessary to be able to control the position of an actuator with precision. Closed-loop control of a motor can be achieved using a position sensor attached to the motor shaft to determine the required control signal to drive it to the desired position. That is the case with a servo motor where a controller drives the motor based on feedback from a position sensor, such as a potentiometer. This mechanism is a key feature distinguishes servo motors from other types of motors, making them a good choice for applications that require accurate and reliable position control.

Introduction

A servomechanism is a device that uses a feedback sensor to automatically adjust the behavior of a system using feedback control. A DC servo motor (often colloquially referred to simply as a servo) is a rotary actuator that transforms electrical energy into mechanical energy and incorporates a feedback mechanism for position control. It holds the commanded position, even under load, until it is instructed to do otherwise. Because they operate in closed-loop control, they can correct disturbances or mechanical load changes by comparing measured versus desired positions.

Basic Construction and Operation

A typical servo consists of four components: a motor (which tends to be a DC motor for hobby servos), a reduction gearbox, a position sensor (a potentiometer or an encoder), and a control circuit. These 4 components can be seen in Figure 1.

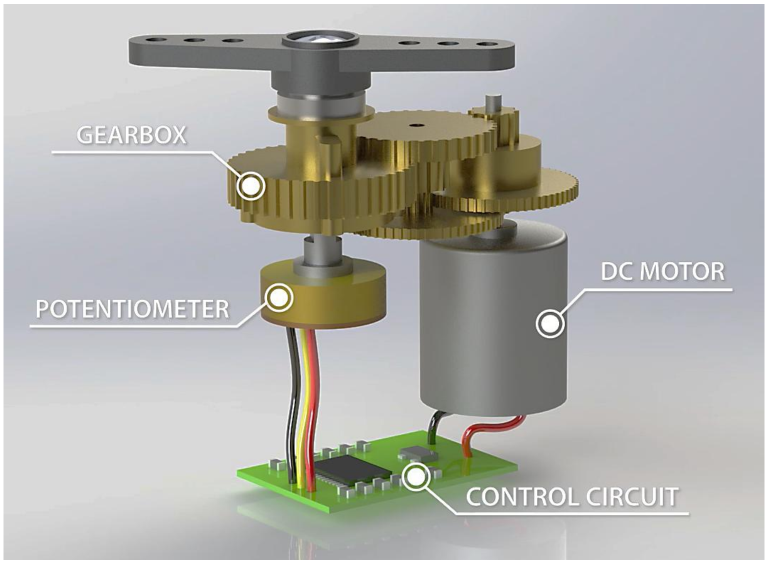

Figure 1. Internal Layout of a Servomotor

The **gearbox** is used to convert the high speed, low torque motor into a high torque, lower speed output. The **potentiometer** measures the current position of the output shaft. The control circuit receives the real time position data from the potentiometer and compares it with the desired position signal. The controller then uses this error between measured and desired position to adjust the input voltage to the motor to move it until the error is minimized and the servo is at the desired position. This allows the servo to keep its position even as load changes dynamically, making them ideal for precise applications. Some systems also use PID control algorithms to optimize accuracy, speed, and stability.

Wiring

Typical wires and connectors of hobby servos are shown in Figure 2. Servos have three cables going into the connector: power, ground and signal. The power cable is typically the middle connector as a safety measure to prevent major damage to the servo if it is connected backwards. The ground wire is usually either brown or black. The signal wire is often orange, but can vary between brands to be yellow, white or even blue. Confirm your servo wiring diagram based on your motor's datasheet.

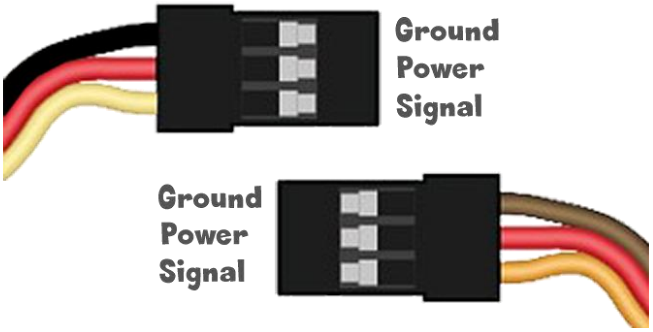

Figure 2. Hobby Servomotors Wiring Connector

Driving a Servomotor

The position of a servo motor is generally commanded using a series of pulses using a pulse width modulation (PWM). A typical servo expects a pulse every 20 milliseconds, which corresponds to a frequency of 50 Hz. The amplitude, or voltage, of this pulse depends on the servo's operating voltage, which tends to be a range which typically includes 5V as part of that range. The pulse width, which is the duration that the signal is high (on), determines the position of the motor.

The pulse width range is usually found within the servo specifications and indicates the minimum and maximum pulse widths corresponding to the minimum and maximum angular positions of the motor shaft. For example, sending a PWM signal with the minimum pulse width will command the motor to move to its minimum angle. The neutral pulse width corresponds to the center position. For a typical hobby servo with a maximum range of 180˚, the minimum pulse is usually a 1 ms pulse for a position at 0˚, the neutral pulse is a 1.5 ms pulse for a position at 90˚, and the maximum pulse is a 2 ms pulse for a 180˚ position. However, the pulse width range could vary depending on the servo. Some servos, like the one provided with the actuators trainer, accepts a pulse between 0.5 ms to 2.5 ms as shown in Figure 3. Note that it is not recommended to operate the servo outside of its pulse range since it can damage or break the servo.

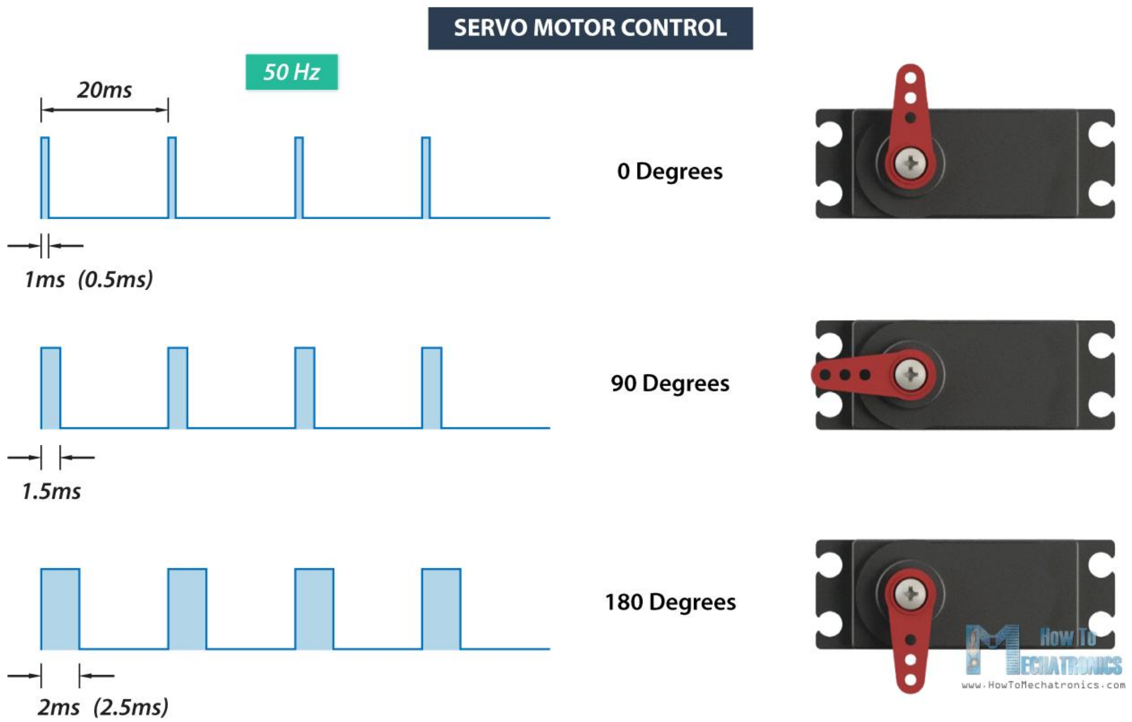

Figure 3. Control Signal for a Servomotor

Datasheet And Considerations

The motor's datasheet summarizes its key electrical and mechanical characteristics and are essential when selecting a motor for a project or application.

Understanding a Servomotor Datasheet

The datasheet lists some properties that should be considered when selecting a servomotor for a project. Figure 4 shows two excerpts from the datasheet for the servomotor provided with the actuators trainer, showing most of the common specifications to consider. Additional specifications are discussed below.

This section will mostly focus on the electrical information, however, note that datasheets also contain mechanical information about the motor, usually in the form of a mechanical drawing. This information is important for deciding if a motor will physically fit in a required space, or mount effectively.

4.电气特性 Electrical Specification					
No	Item	Specifcation			
4.1	工作电压 Operat voltage	4.8V	6.0V	7.2V	8.4V
4.2	静态电流 Idle current	＜40mA			
4.3	空载转速 No load speed	0.10sec/60°	0.09sec/60°	0.08sec/60°	0.07sec/60°
4.4	空载电流 Runnig current	210mA	260mA	350mA	450mA
4.5	堵转扭矩 Peak stall torque	12kg.cm	14.5kg.cm	15.5kg.cm	16.5kg.cm
4.6	堵转电流 Stall current	2700mA±10%	3200mA±10%	3900mA±10%	4300mA±10%

5.控制特性 Control Specification		
No	Item	Specifcation
5.1	控制信号 Command signal	Pulse width modification
5.2	放大器类型 Amplifier type	Digital controller
5.3	脉冲宽度范围 Pulse width range	500～2500usec
5.4	中点位置 Neutral position	1500usec
5.5	旋转角度 Running degree	180°±3°(when 500～2500usec)
5.6	死区宽度 Dead band width	3 usec

Figure 4. Servomotor Datasheet Excerpt

Rated Voltage

Also listed as the operating voltage, his is the voltage at which the motor is designed to run, and at which the motor will have been tested for other specifications. If the supply voltage is less than the operating voltage, the system will be unable to produce the rated torque. If the supply voltage exceeds the rated voltage, there is a risk of damage to the motor hardware. Most hobby servos operate at 4.8–7.4 V, but industrial versions can handle higher ranges.

Peak Current

This is the current which the servo motor will draw, which occurs when it is exerting peak torque. This will occur when commanding a large step in angle, or when the servo is resisting a large external torque. Current draw varies with load and activity—peak current occurs during acceleration or when resisting force.

Max/Holding Torque

The holding torque is determined by several factors, such as the gearbox and the DC motor properties. Most applications will have a torque requirement either to actuate a particular mechanical assembly, or to hold position in the face of a specified external load. This determines the highest load the servo should be able to resist or apply. Exceeding it causes position errors, but the servo will typically return to its target once the load is removed.

Servo motors are usually rated in kilograms per centimeter (kg/cm), and most hobby servos like the SG90, MG995 etc, are rated between 3kg/cm to 12kg/cm. This rating indicates the weight the motor can lift at a specific distance. For example, a 25kg/cm servo can lift 25kg when the load is 1cm from the motor shaft as shown in Figure 5. The maximum load at each position is the torque divided by the distance in cm.

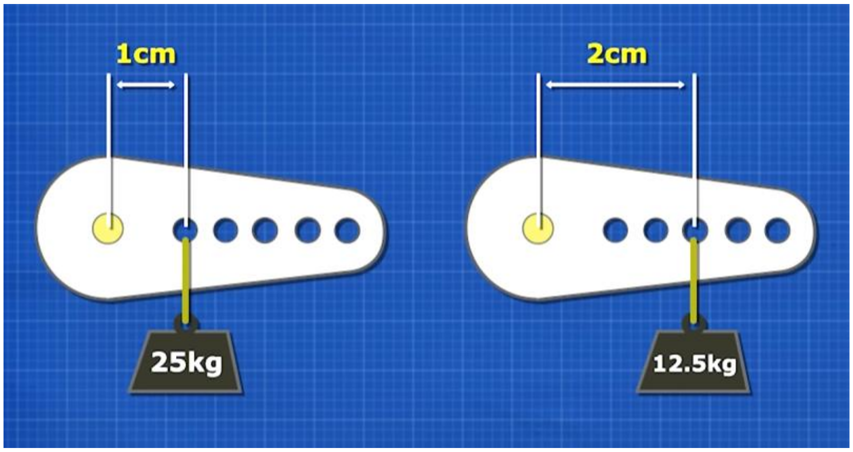

Figure 5. Motor Torque measured in kg/cm

Servo Speed

Servo speed is usually given as the time to move a fixed angle (e.g., 0.1 s per 60°) at a specific voltage. This gives a more practical measure of responsiveness compared to RPM.

Range

The range of a servo motor refers to the minimum and maximum angle it can rotate from its neutral position. Most hobby servos provide a range of 180°, although some models support extended motion up to 270°, or even continuous rotation. The required range depends on the mechanical setup—some applications only need limited angular motion, while others benefit from wider or unrestricted travel.

Resolution and Deadband

The resolution indicates the smallest change in position of the shaft that can be achieved by the servo. The deadband indicates the smallest input signal that is required to produce a change in the output of the servo. Both specifications indicate limits on the accuracy and precision of control that can be achieved by the motor.

Design Decisions

Cost

The final cost of a motor will depend on variables including volume of production, and the location of the manufacturer. Hobby servos are low-cost and widely available, while industrial-grade models are more expensive due to higher torque, precision, and durability.

Efficiency

Servo motors are extremely efficient under low or no load, only drawing power when correcting position. However, they may heat up when subjected to high variable torques as the control circuitry must constantly drive high currents through the motor coils to resist the applied torque.

Ease of Use

Most servos are simple to use with standard PWM signals and require no external control circuitry. More advanced models may use digital protocols and require configuration.

Lifespan

The estimated lifespan of the motor is listed either in hours of operation or number of rotations. Servo components include brushed DC motors or plastic gears that wear over time. Lifespan varies based on use, load, and component quality.

Range of Motion

While this document focused mainly on servo motors with a limited range of motion, there are also free-turning or continuous servos available on the market. However, for continuous motion in one direction, DC motors are a more robust and cost-effective choice for a given torque and power requirement, omitting the extra components needed for positional control in a servo.

Position Control Accuracy

This is an area where the servo motor has significant advantages over other motors. Servos provide reliable position control through internal feedback. Gear reduction and PID control ensure good repeatability, even under varying loads without the loss of torque.